

TEST ON PAINTS

IS 101

main tests on paints used in construction

1. **Viscosity Test –**
2. **Drying Time Test –**
3. **Adhesion Test –**
4. **Opacity (Hiding Power) Test –.**
5. **Gloss Test –**
6. **Hardness Test –**
7. **Water Resistance Test –**
8. **Chemical Resistance Test –**
9. **Flexibility Test**
10. **Abrasion Resistance Test**
11. **Weathering Test**
12. **Film Thickness Test**

Tests on Paints for Construction

Evaluating Paint Quality & Performance

 Viscosity Test Flow & Thickness	 Water Resistance Test Moisture Protection
 Drying Time Test Time to Dry	 Chemical Resistance Test Acids & Alkalis
 Adhesion Test Bond to Surface	 Flexibility Test Bend Without Cracking
 Opacity Test Hiding Power	 Abrasion Test Wear Resistance
 Gloss Test Surface Shine	 Weathering Test Sun, Rain & Heat
 Hardness Test Scratch Resistance	 Film Thickness Test Coating Thickness

As Per IS 101 – Methods of Test for Paints & Varnishes

Viscosity Test on Paints

The **Viscosity Test** is used to determine the **flow and thickness of paint**. It ensures that the paint has the correct consistency for **proper application on surfaces like walls, metal, or wood**.

Standard

- As per **IS 101 – Methods of Test for Paints and Varnishes** published by the Bureau of Indian Standards.

Apparatus

- Viscometer (Ford Cup / Stormer Viscometer)
- Stopwatch
- Paint sample container
- Thermometer

Procedure

1. Fill the **viscometer cup** with the paint sample.
2. Allow the paint to **flow through the orifice** of the cup.
3. Start the **stopwatch** when the paint begins to flow.
4. Stop timing when the flow **breaks or stops**.
5. Record the **time in seconds**.

Result / Measurement

- Viscosity is usually expressed in **seconds or centipoise (cP)**.
- Higher time → **Thicker paint**
- Lower time → **Thinner paint**

Purpose

- Ensures **uniform paint application**
- Helps in **proper spreading and leveling**
- Prevents **dripping or uneven coating**

Viscosity Test in Paints

Measuring the *Flow & Thickness* of Paint



Specifications

- Tool Used: Viscometer
- Standard: IS 101 (Part 2)
- Checks: Flow, Consistency & Thickness
- Measurement: In Centipoises (cP)



- ✓ Ensures Proper Application
- ✓ Testing for Sprayability
- ✓ Uniform Coverage of Paint



- ✓ Ensures Proper Application
- ✓ Testing for Sprayability
- ✓ Uniform Coverage of Paint

Drying Time Test on Paints

The **Drying Time Test** is performed to determine the **time required for paint to dry after application** on a surface.

Standard

- As per **IS 101 – Methods of Test for Paints and Varnishes** by the Bureau of Indian Standards.

Apparatus

- Paint sample
- Glass panel or metal plate
- Brush or applicator
- Stopwatch
- Drying time recorder (optional)

Procedure

1. Apply a **uniform coat of paint** on a clean glass or metal panel.
2. Keep the panel under **standard laboratory conditions**.
3. Check the paint film at regular intervals.
4. Record the time for different stages of drying.

Types of Drying Time

1. **Surface Dry Time** – Time when the paint surface becomes **touch dry**.
2. **Hard Dry Time** – Time when the paint film becomes **completely hard and dry**.

Purpose

- Ensures **quick and proper drying of paint.**
- Helps in **efficient painting work** in construction.
- Prevents **dust sticking and surface damage.**

Typical Specification

- Surface dry: **30 minutes**



Adhesion Test on Paints

The **Adhesion Test** is carried out to determine how well the **paint film sticks to the surface** such as concrete, plaster, metal, or wood.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Cross-cut adhesion tester or sharp blade
- Adhesive tape
- Magnifying glass
- Painted test panel

Procedure (Cross-Cut Tape Test)

1. Apply paint on a **clean test surface** and allow it to dry completely.
2. Make **small square cuts (grid pattern)** on the paint film using a blade or cross-cut tester.
3. Apply **adhesive tape** over the cuts.
4. Pull the tape **quickly at a 180° angle**.
5. Observe how much **paint film is removed** from the surface.

Result / Rating

Adhesion is rated based on the **amount of paint removed**:

- **5B / Excellent** – No paint removed
- **4B / Good** – Very small flakes removed
- **3B / Moderate** – Some coating removed
- **2B–0B / Poor** – Large amount of paint removed

Purpose

- Ensures **strong bonding** of paint to the surface
- Prevents **peeling, flaking, and blistering**
- Improves **durability** of paint coating

Adhesion Test on Paints

Measuring How Well Paint Sticks to Surfaces



Specifications

- **Standard:** IS 101 (Part 6)
- **Checks:** ✓ Concrete
✓ Plaster
✓ Metal
✓ Wood
- **Rated from:** 0 (Poor) to 5 (Excellent)



- ✓ Ensures **Strong Bond** to Surface
- ✓ Prevents **Peeling** & Chipping
- ✓ Checks Paint Durability

Opacity (Hiding Power) Test on Paints

The **Opacity Test** is carried out to determine the **hiding power of paint**, i.e., the ability of the paint to **cover the background surface completely**.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Black and white test chart (opacity chart)
- Brush or applicator
- Paint sample
- Reflectance meter (optional)

Procedure

1. Take a **black and white contrast chart**.
2. Apply a **uniform coat of paint** on the chart.
3. Allow the paint to **dry completely**.
4. Observe whether the **black and white areas are visible** through the paint film.

Result

- If the background **cannot be seen**, the paint has **high opacity (good hiding power)**.
- If the background **is still visible**, the paint has **low opacity**.

Purpose

- Ensures **better surface coverage**
- Reduces the **number of paint coats required**
- Improves the **appearance and finish of painted surfaces.**



Gloss Test on Paints

The **Gloss Test** is conducted to determine the **shine or reflective property of a painted surface**. It indicates how much **light is reflected from the paint film**.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- **Gloss meter**
- Painted test panel
- Light source

Procedure

1. Apply a **uniform coat of paint** on a clean test panel.
2. Allow the paint film to **dry completely**.
3. Place the **gloss meter** on the painted surface.
4. Measure the **amount of light reflected** from the surface.

Result

- The gloss value is measured in **Gloss Units (GU)**.
- Higher value → **High gloss (more shiny surface)**
- Lower value → **Low gloss (matte surface)**

Purpose

- Determines the **appearance and finish of paint**.
- Ensures **uniform shine on painted surfaces**.
- Helps in selecting **matte, semi-gloss, or high-gloss paint** for construction.

Gloss Test on Paints

Measuring the Shine & Reflective Quality of Paint



Glossy Paint

Matte Paint



Gloss Meter

Specifications

- Tool Used: Gloss Meter
- Standard: IS 101 (Part 5)
- Measures: Surface Reflectivity
- Unit: Gloss Units (GU)





Matte Semi-Gloss Gloss High-Gloss

- ✓ Ensures **Desired** Shine Level
- ✓ Better Aesthetic **Appeal**
- ✓ Testing for **Paint Quality**

Hardness Test on Paints

The **Hardness Test** is performed to determine the **resistance of the dried paint film to scratching, abrasion, or mechanical damage**.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Pencil hardness tester / Scratch tester
- Set of graded pencils (6B to 9H)
- Painted test panel

Procedure

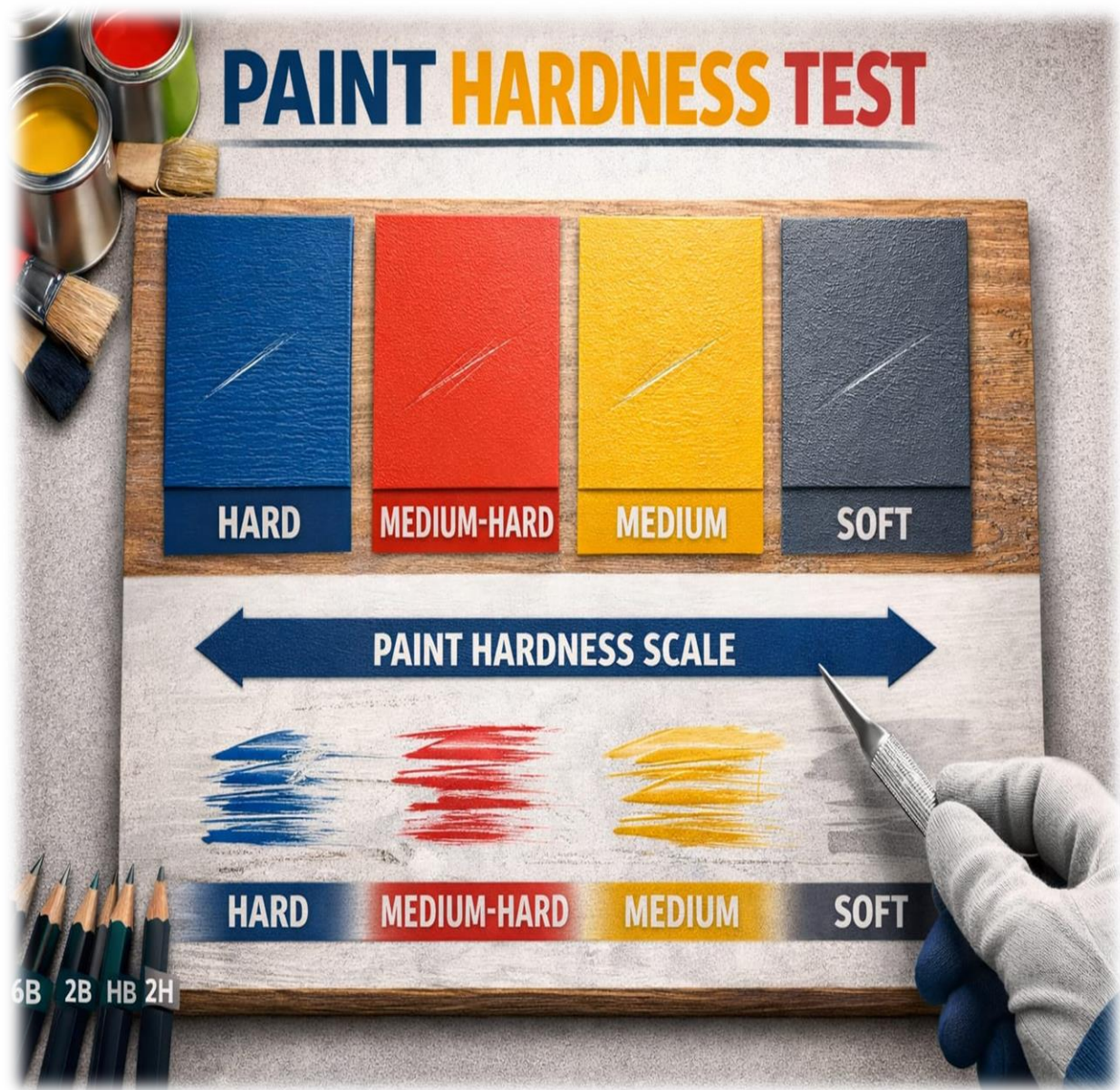
1. Apply a **uniform coat of paint** on a clean test panel and allow it to **dry completely**.
2. Hold the **pencil at about 45° angle** to the painted surface.
3. Push the pencil across the surface with **uniform pressure**.
4. Observe whether the paint film **scratches or remains intact**.

Result

- The hardness is expressed by the **hardest pencil that does not scratch the paint film**.
- Higher hardness → **Better resistance to scratches and wear**.

Purpose

- Ensures **durability** of the paint coating.
- Prevents **surface damage** due to friction or impact.
- Improves the **service life** of painted surfaces.



Water Resistance Test of Paints

The **Water Resistance Test** is conducted to determine the **ability of paint to resist the effect of water or moisture** without damage such as blistering, peeling, or loss of adhesion.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Painted test panel (metal or glass)
- Water container or immersion tank
- Stopwatch
- Dry cloth

Procedure

1. Apply a **uniform coat of paint** on the test panel and allow it to **dry completely**.
2. Immerse the painted panel in **water for a specified period** (usually 24 hours).
3. Remove the panel from the water.
4. Observe the paint film for **blistering, peeling, softening, or discoloration**.

Result

- If the paint film **remains intact**, it has **good water resistance**.
- If the paint film **blisters or peels**, the water resistance is **poor**.

Purpose

- Ensures **paint durability** in humid or wet conditions.
- Protects surfaces from **moisture damage**.
- Important for **exterior walls, bathrooms, and kitchens**.



Chemical Resistance Test of Paints

The **Chemical Resistance Test** is conducted to determine the **ability of the paint film to resist chemicals such as acids, alkalis, oils, and solvents** without damage.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.
- Painted test panel (metal or glass plate)
- Chemical solutions (acid, alkali, or solvent)
- Dropper or brush
- Stopwatch
- Clean cloth

Procedure

1. Apply a **uniform coat of paint** on the test panel and allow it to **dry completely**.
2. Place **a few drops of chemical solution** (acid or alkali) on the painted surface.
3. Leave it for a **specified time (usually 24 hours)**.
4. Clean the surface and **observe the paint film**.

Result

- If the paint film shows **no change**, it has **good chemical resistance**.
- If the paint film shows **blistering, discoloration, or peeling**, the resistance is **poor**.

Purpose

- Ensures the paint can **withstand chemical exposure**.
- Protects surfaces in **industrial areas, laboratories, kitchens, and factories**.
- Improves the **durability and life of the coating**.

Chemical Resistance Test on Paints

Checking Resistance to Chemicals, Acids & Alkalis



Specifications

- **Tool Used:** Test Panels
- **Standard:** IS 101 (Part 8)
- **Checks:** Resistance to Acids
- **Testing:** ☒ Resistance to Alkalis
☒ Resistance to Acids
☒ Exposure Time & Surface Effects



Gloss Meter

- ✓ **Durability Against Chemicals**
- ✓ **Prevents Paint Deterioration**
- ✓ **Ensures Long-Lasting Protection**

Flexibility Test of Paints

The **Flexibility Test** is conducted to determine the **ability of a paint film to bend or stretch without cracking or peeling** when the surface is deformed.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Mandrel bending tester or cylindrical rod
- Painted metal panel
- Magnifying glass

Procedure

1. Apply a **uniform coat of paint** on a metal panel.
2. Allow the paint film to **dry completely**.
3. Bend the painted panel **around a cylindrical rod (mandrel)**.
4. Observe the paint film for **cracks, peeling, or damage**.

Result

- If **no cracks appear**, the paint has **good flexibility**.
- If **cracks or peeling occur**, the paint has **poor flexibility**.

Purpose

- Ensures the paint film can **withstand expansion and contraction of surfaces**.
- Prevents **cracking of paint coatings**.
- Improves the **durability of painted structures**.

Flexibility Test on Paints

Measuring If Paint Bends Without Cracking or Flaking



Flexible



Cracked

Specifications

- **Tool Used:** Cylindrical Mandrel
- **Standard:** IS 101 (Part 10)
- **Checks:** Paint's Resistance to Bending:



Testing Method:
Cylindrical Mandrel Bend Test



IS 101
Part 10

- ✓ Ensures **Paint Will Not Crack**
- ✓ Prevents **Flaking & Chipping**
- ✓ Testing **Paint Durability**

Abrasion Resistance Test of Paints

The **Abrasion Test** is conducted to determine the **ability of a paint film to resist wear, rubbing, or friction** during service.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Abrasion testing machine (Taber Abraser)
- Painted test panel
- Abrasive wheels
- Weight load

Procedure

1. Apply a **uniform coat of paint** on the test panel and allow it to **dry completely**.
2. Place the panel in the **abrasion testing machine**.
3. Rotate the **abrasive wheels over the paint surface** under a specified load.
4. Run the machine for a **fixed number of cycles**.
5. Observe the **amount of paint wear or loss of film thickness**.

Result

- **Less wear of paint film** → Good abrasion resistance.
- **More wear or damage** → Poor abrasion resistance.

Purpose

- Ensures the paint can **withstand rubbing, cleaning, and mechanical wear**.
- Important for **floors, walls, doors, and high-traffic areas**.
- Improves the **durability and service life of paint coatings**.

Abrasion Resistance Tests on Paints

Acrylic Paint	Enamel Paint	Low-Quality Paint
✓	✓	✗
		
✓ No Visible Damage ✓ High Durability	● Light Wear ● Moderate Durability	✗ Heavy Scuffing ✗ Poor Durability

1,000 Cycles Scrubbing Test



Abrasion Resistance Test Method Using a 2.5 lbs weighted abrasive scrubber with 1,000 scrubbing cycles using 3,000 grit sandpaper.

 1,000 Scrubbing Cycles	 Grit: 3,000	 Pressure:	 2.5 lbs (1.13 kg) Pressure: 2.5 lbs (1.13 kg)
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Weather Resistance Test of Paints

The **Weather Resistance Test** is carried out to determine the **ability of paint to withstand weather conditions** such as **sunlight, rain, temperature changes, and humidity** without deterioration.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Painted test panel (metal or wood)
- Weathering chamber or outdoor exposure rack
- UV light source (in laboratory tests)
- Observation tools

Procedure

1. Apply a **uniform coat of paint** on a clean test panel.
2. Allow the paint film to **dry completely**.
3. Expose the panel to **natural weather conditions** or in a **weathering chamber** with UV light and moisture.
4. Observe the panel after a **specified period** for any defects.

Result

- If the paint film shows **no cracking, fading, or peeling**, it has **good weather resistance**.
- If defects appear, the paint has **poor weather resistance**.

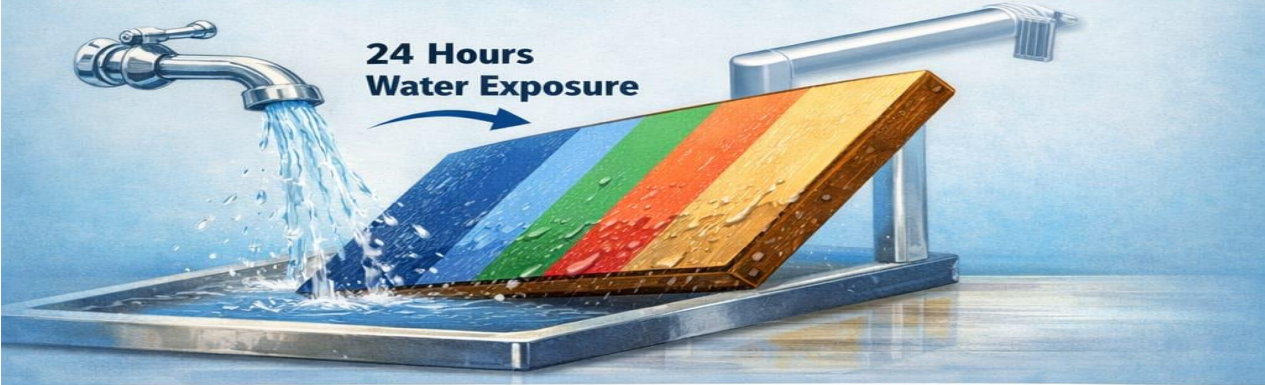
Purpose

- Ensures paint can **withstand outdoor environmental conditions**.
- Protects buildings from **sunlight, rain, and humidity**.
- Improves the **durability and life of exterior paint coatings**.

Water Resistance Test on Paints

Acrylic Paint	Enamel Paint	Low-Quality Paint
		
• Water Beading • No Paint Damage	• Minor Water Spots • Slight Surface Wear	• Peeling & Blistering • Paint Dissolving

24 Hours Water Exposure



Test Duration: 24 Hours

Temp: 25°C (77°F)

Condition: Constant Water Flow

Film Thickness Test on Paints

The **Film Thickness Test** is carried out to determine the **thickness of the dried paint coating** on a surface. Proper thickness ensures **good protection, durability, and appearance** of the paint.

Standard

- Conducted according to **IS 101 – Methods of Test for Paints and Varnishes** issued by the Bureau of Indian Standards.

Apparatus

- Film thickness gauge (Dry Film Thickness Gauge)
- Painted test panel
- Micrometer or electronic thickness meter

Procedure

1. Apply a **uniform coat of paint** on the test surface.
2. Allow the paint to **dry completely**.
3. Place the **film thickness gauge** on the painted surface.
4. Measure the **thickness of the paint film** at different points.

Result

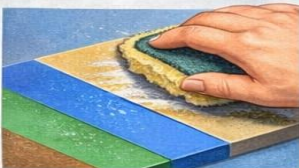
- The thickness is measured in **microns (μm)** or **millimeters (mm)**.
- If the thickness is **within the specified range**, the coating is acceptable.

Purpose

- Ensures **adequate protection** of the surface.
- Prevents **excess or insufficient paint coating**.
- Improves the **durability and performance of paint**.



Film Tests on Paints Before Using

Adhesion Test  ✓ Assessing Adhesion Strength	Flexibility Test  ✓ Testing Paint Flexibility	UV Resistance Test  ✓ Evaluating UV Durability
Scrubbing Test  ✓ Measuring Abrasion Resistance	Crack Resistance Test  ✓ Measuring Abrasion Resistance	Crack Resistance Test  ✓ Checking for Cracks & Peeling

Ensure Paint Quality Before Application